

|                                                                                   |                                                                                                                                                                                                 |                                                  |                        |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|------------------------|
|  | Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.<br>Department of Chemical Engineering<br>In-Semester Examination II<br>Academic Year: 2025-26 (semester I) |                                                  |                        |
|                                                                                   | Class: Second Year                                                                                                                                                                              | Course: Chemical Engineering<br>Thermodynamics-I | Course Code: PCC-CH203 |
|                                                                                   | Date: 13.10.2025                                                                                                                                                                                | Time: 14:30 to 15:15                             | Max. Marks: 20         |
|                                                                                   |                                                                                                                                                                                                 |                                                  |                        |

Note:

- Assume suitable data whenever necessary and state it clearly.
- Figures to the right indicate full marks.

| Q.N. | Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Mark | CO | BT  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----|-----|
| Q.1  | (a) Draw the compressibility chart for the industrial gases and state the principle of corresponding state.<br>(b) Define entropy and derive the expression for 'entropy change for a reversible process in isothermal process involving ideal gas<br>(c) State the second law of thermodynamics statement and write mathematical expression for the same.                                                                                                                                                                                                                                                                                   | 4    | 3  | 2   |
| Q.2  | (a) Calculate the change in internal energy, change in enthalpy, work done, and the heat supplied in the following processes:<br>(i) An ideal gas is expanded from 5 bar to 4 bar isothermally at 600 K.<br>(ii) An ideal gas contained in a vessel of 0.1 m <sup>3</sup> capacity is initially at 1 bar and 298 K. It is heated at constant volume to 400 K.<br>Assume that $C_p = 30 \text{ J/mol K}$ .                                                                                                                                                                                                                                    | 4    | 2  | 1&6 |
| Q.3  | (a) An ideal gas is undergoing a series of three operations. The gas is heated at constant volume from 300 K and 1 bar to a pressure of 2 bar. It is expanded in a reversible adiabatic process to a pressure of 1 bar. It is cooled at a constant pressure of 1 bar to 300 K. Determine the heat and work effects for each step. Assume $C_p = 29.3 \text{ J/mol K}$ .<br><br>(a) With neat sketch list out the various features of P-V-T behaviour for pure fluid for the variation of Pressure Vs molar volume.<br><br>(a) Explain the Carnot cycle with a neat labelled P-V diagram and differentiate between heat pump and heat engine. | 6    | 2  | 5   |
|      | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4    | 3  | 2&4 |
|      | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 4    | 3  | 2&4 |

=== ALL THE BEST ===

Bloom's Taxonomy (BT)  
BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

#### Course Outcomes:

- On successful completion of this course, students should be able to:
- CO 1 Able to know the fundamental concepts of thermodynamics.
  - CO 2 Able to apply the first and second law of thermodynamics for chemical processes.
  - CO 3 Able to relate PVT relations and equation of state for pure component systems.
  - CO 4 Evaluate thermodynamic properties in chemical engineering applications.
  - CO 5 Understand the application of thermodynamics to various flow processes also of refrigeration and liquefaction